

Digitalized health workforce education: an elicitation of research gaps and selection of case studies



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WHO Collaborating Centre for
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Abbreviations

BemONG	basic emergency obstetrics and newborn care
BHPMx	Better Health Programme Mexico
DELIVER	Digital Educational programme involVing hEalth pROfessionals
DHA	Dubai Health Authority
DHWE	digitalized health workforce education
EU	European Union
HCP	health-care professional
mLearning	mobile learning
RIEC	Research, Innovation and Enterprise Council (Singapore)
SBA	skilled birth attendant
SDA	Safe Delivery App
USA	United States of America
WHO	World Health Organization

Executive summary

The third goal of the United Nations Sustainable Development Agenda aims to ensure that everyone has access to good health and well-being by 2030. To achieve this, countries must establish responsive and people-centred health systems, as well as an efficient, supported and well-motivated health workforce. The COVID-19 pandemic has further demonstrated the significance of the health workforce and the importance of investing in solutions that address the challenges it faces. To address these challenges adequately, policy- and decision-makers require evidence-based guidance.

Recently, the availability of technological learning tools has experienced exponential growth, significantly altering the traditional way health workers have been educated and how they deliver health services. The benefits and limitations of using digital tools and platforms to supplement traditional methods of educating health workers have been highlighted in several studies to date. Findings show that the effectiveness (i.e. what does or does not work) of using digital technology to educate and train health workers varies according to training objectives, digital modality, context, teaching and assessment methods, study population and specialty of practice.

This report presents a broad summary of the research gaps around the use of digital technology for the education and training of health workers. Additionally, it applies a conceptual framework developed from a synthesis of existing frameworks for digital health in general, as well as digitalized health workforce education (DHWE). The report includes a non-exhaustive, non-prioritized list of 63 research questions, which are classified across six domains of the applied conceptual framework to guide future research in advancing DHWE. Finally, evidence gaps from the literature are identified to inform future research activities.

1. Introduction

According to the latest WHO projections, a shortage of an estimated 10.2 million health workers is expected by 2030.¹ The challenges posed by health worker shortages are more exacerbated in low- and middle-, than in high-income countries in a variety of ways, including unequal distribution of labour, working conditions, wages and job requirements. Taken together, these problems jeopardize countries' chances of achieving the third goal of the United Nations Sustainable Development Agenda i.e. that everyone has access to good health and well-being by 2030.² WHO, its Member States, and partners have invested significant resources in providing normative guidance, technical support and other forms of assistance to help countries address these challenges and achieve universal health coverage.^{3,4}

Technology plays an important role in helping countries to do this and to deliver effective health services.⁵ To meet the increasing demand for health services, it is essential to explore the role of education and training in a labour market approach to develop a fit-for-purpose and sustainable health workforce.⁶ Under these conditions, there are opportunities to increase the digitization of health worker education to facilitate accessibility, increase gains and support lifelong learning.⁷⁻⁹

Technological advances have transformed the educational landscape of the health workforce and the nature of the services they deliver.^{7,10} Thus, health workers and their educators need to improve their respective understanding of how digital tools can be most effectively integrated into the education and lifelong learning of health workers.¹¹ Digitalized health workforce education (DHWE), also referred to as digital health professions education, is simply defined as the use of digital tools and platforms to deliver education or training to learners, whether they are pre- or in-service health workers. It is still debatable, however, for which groups of learners complete DHWE is self-sufficient, and for which groups it should be combined with traditional methods, given the clinical nature of many health workforce occupational roles.

Supplementing health workforce education with digital tools and resources requires that policy-makers be aware of what pedagogy, educational design, effectiveness, implementation and resources they need to ensure success. Until recently, the use of digital technologies in health workforce education has mainly consisted of supporting face-to-face instruction through the digitization of learning materials and the use of a digital trainer to enhance psychomotor skills. However, current trends point to a diversification of technological modalities to include mobile learning (mLearning), digital game-based learning, and virtual patients among others.^{12,13} Digital tools and resources are also increasingly being used for distance education, extending the reach of education to health workers located in rural and remote areas.¹⁴

The evidence on DHWE has significantly grown, with extant reviews mostly centred on the effectiveness of diverse DHWE modalities.^{15–18} To provide comprehensive guidance on future research, there is a pressing need to systematically identify the available evidence and research gaps. This report aims to articulate the gaps in the existing literature in a way that offers a more comprehensive insight and targets decision-makers, educators, researchers and curators of research and learning programmes. The goals are:

1. to provide educators and researchers in countries and relevant institutions with a list of non-prioritized key questions to consider when planning research activities on digital tools and resources for the education of health workers;
2. to provide examples and lessons to guide capacity development for DHWE research; and
3. to identify gaps in the existing literature to highlight areas for future research.

2. Methods

This report is based on a published review conducted by Tudor Car et al.¹⁹ to map existing evidence in DHWE. The researchers searched multiple databases and grey literature sources from January 2014 to July 2020 for systematic reviews on DHWE for practising health-care professionals and students in training. After outlining the quality of the synthesized evidence, this evidence was then mapped against a newly developed conceptual framework. To ensure the report's relevance and currency, we conducted an additional search covering the period up to June 2022. The evidence from Tudor Car et al.'s review and the new search was then analysed using an evidence map methodology²⁰ to derive research questions that corresponded to the developed conceptual framework described by Tudor Car et al. Evidence mapping is a systematic method for identifying and categorizing existing literature on a specific topic. The goal of an evidence map is to provide a broad overview of the existing literature on a topic while also identifying knowledge gaps and/or future research needs. Evidence mapping is frequently used as a preliminary step in determining future research priorities or conducting more comprehensive systematic reviews. Case studies were selected if they included critical success factors that are instrumental to promote research for DHWE. The case studies were identified using several iterations of topical google searches related to DHWE, including reviewed published best practice examples known to the authors of this report.

2.1 Identifying relevant studies, data extraction and analysis

The following databases were part of the initial search for systematic reviews on the digital education of practising and student HCPs: CINAHL, Cochrane Library, Educational Research Information Center (ERIC), Embase, MEDLINE, and grey literature sources. Search terms were the same as applied in the aforementioned study by Tudor Car et al.¹⁹ Other databases searched were Google, Google Scholar, ResearchGate, and OpenGrey. For the latter databases, the following keywords were applied: *systematic review*, *digital education*, *health professionals*, *health professions education*, *eLearning*, and *e-learning*, and the first 10 pages or 500 results were included for review. Following the terms, database search and the extraction of relevant studies, 77 eligible systematic reviews that evaluated the effectiveness of DHWE interventions were included for analysis. The period of studies included was from 1 January 2014 to 21 July 2020.

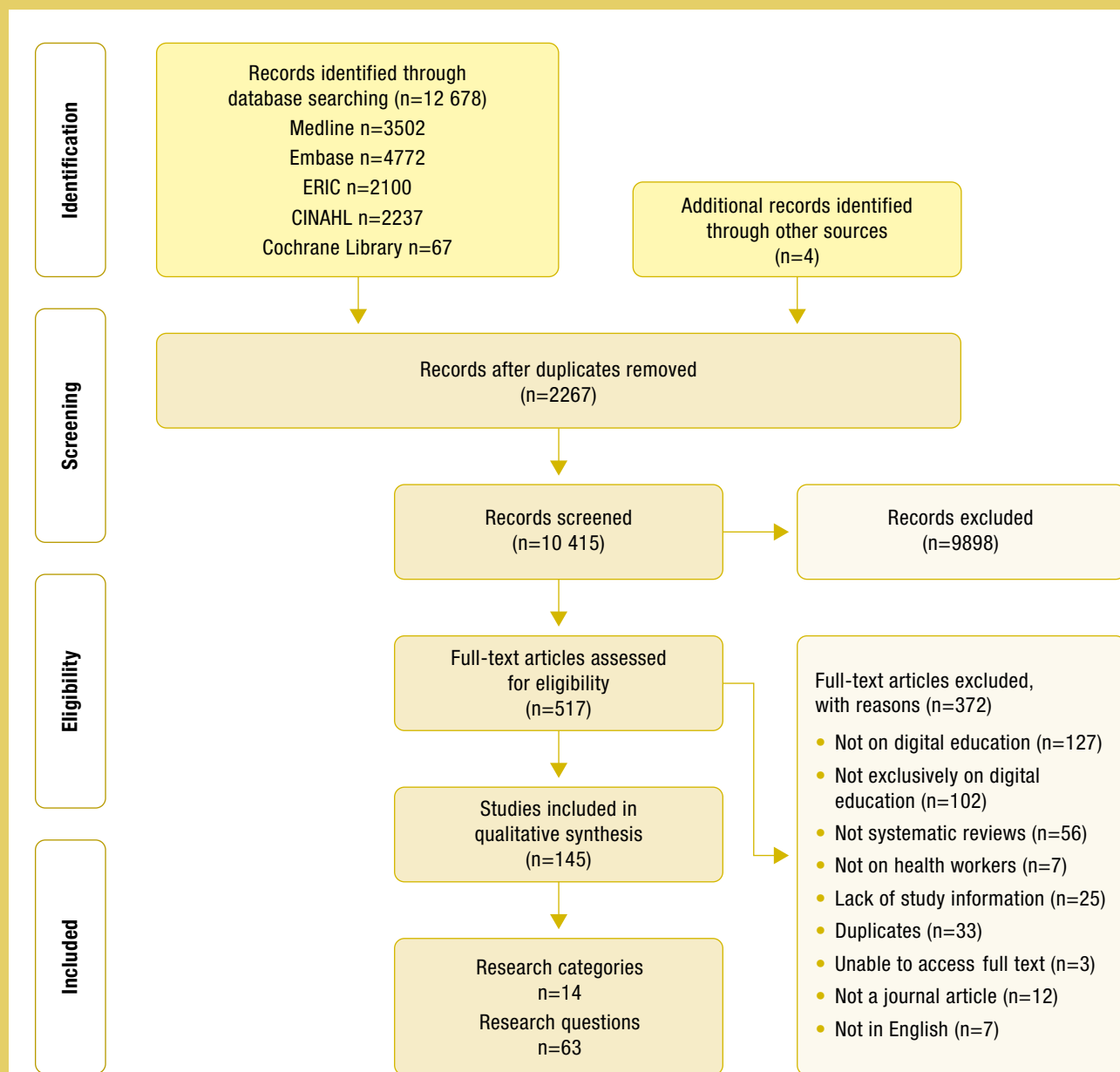
A second round of searching was conducted to update the findings and identify new, more recent systematic reviews using the same search terms. The same databases were searched again with the date period of 22 July 2020 to 30 June 2022. An additional 68 systematic reviews were deemed eligible and included in the data analysis.

Systematic reviews were included if they focused on one or more modalities of DHWE delivered in a stand-alone or blended format. Following data extraction, digital modalities were categorized in accordance with the classification used in each respective systematic review to specify the digital modality. Relevant statements related to the research recommendations were extracted from the discussion sections of the systematic reviews and rephrased into research questions. To ensure high-quality and relevant data, several exclusion criteria were applied. Studies that did not focus on DHWE (i.e. both stand-alone and blended formats) or that were not exclusively for education (i.e., digital modalities used for non-education purposes) were excluded. Furthermore, studies that did not meet the criteria for systematic reviews, were duplicates, or lacked adequate information were excluded. Publications that were either irrelevant to health professionals or did not address our research questions were also excluded. Finally, studies that could not be accessed in full, were not in English, or had the wrong publication type (e.g. conference abstracts, proposals, editorials, commentaries, letters to editors, preprints, and short letters articles), were excluded.

3. Findings

Initial search results (including both rounds of searching) identified 12 678 systematic reviews. Four additional systematic reviews were identified through the other databases search (Fig. 1). In total, 145 (1%) systematic reviews were included for data extraction (see Annex 1 for full details of the 145 reviews).

Figure 1. Study flow diagram*



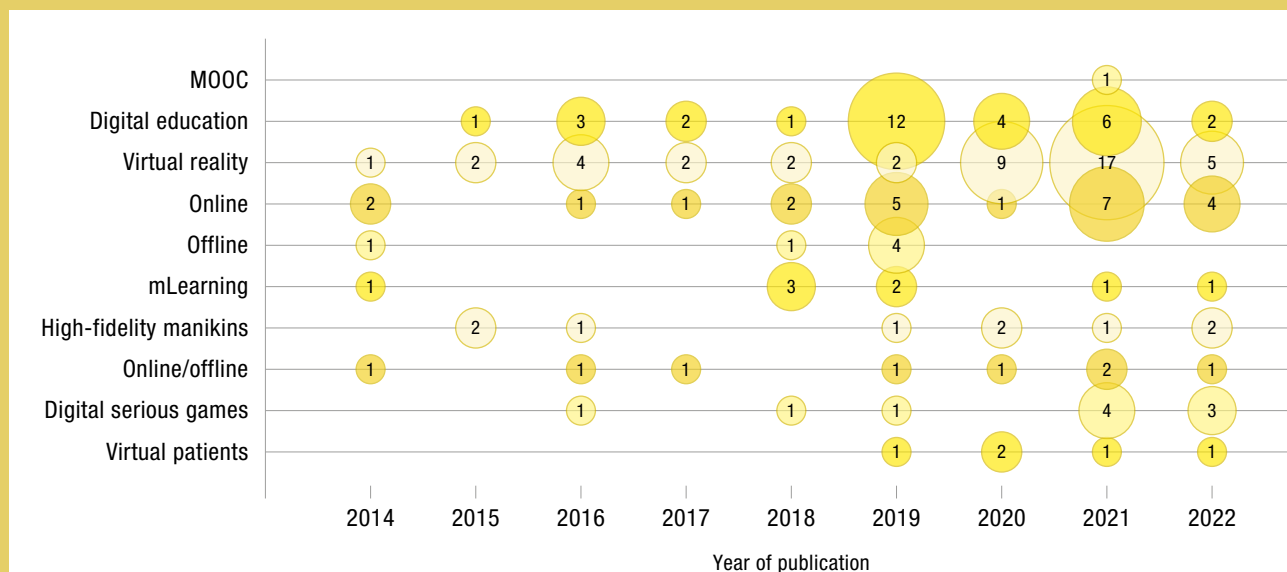
*Based on Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA).

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group. Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement.

For more information, visit www.prisma-statement.org.

Of the 145 systematic reviews included, most focused on DHWE in general (31, 21%), virtual reality (44, 30%) and online education (23, 16%). There were fewer reviews published on mLearning (8, 6%), digital game-based learning (10, 7%) and virtual patients (5, 3%). Of the 44 reviews on virtual reality, 36 (82%) were complemented with physical objects or devices such as probes or handles, and focused on psychomotor, procedural or technical skill development. (Fig. 2).

Figure 2. Number of systematic reviews on different DHWE modalities, January 2014 to June 2022



MOOC: massive open online courses.

The 145 systematic reviews primarily collected data from high- and middle-income countries (70, 48%). This was followed by the studies from only high-income countries (43, 30%), high-, middle-, and low-income countries (10, 7%), middle- and low-income countries (1/145, 1%) and middle-income countries (1/145, 1%). Twenty (14%) of the systematic reviews did not indicate the countries where the studies were conducted.

Most of the 145 included studies focused on mixed groups of participants, such as medical students and physicians (47, 32%), nursing students and nurses (30, 21%), and students and HCPs from a range of health-care specialties and occupations (35, 24%). Eight (6%) studies focused on dentistry students and dentists, and only two (1%) focused on pharmacy students and pharmacists. The remaining studies had only HCPs (14, 10%) and students (9, 6%) as participants.

When delivering DHWE, the studies mostly covered the general knowledge of health professionals (38, 26%) and other core competencies needed for HCPs (40, 28%). The remaining studies covered specific medical specialties or subjects such as nursing (29, 20%), surgery (24, 17%), radiology (7, 5%), endoscopy (3, 2%), and life support or trauma management (4, 3%).

The 145 studies also showed that competency outcomes of knowledge (98, 68%) and skills (91, 63%) were the most commonly assessed. Some studies assessed clinical attitudes (23, 16%) and satisfaction with DHWE (38, 26%). Nonetheless, most reviews only reported immediate, short-term outcomes (115; 79%) while the remaining 30 (21%) reported on the impact of DHWE on longer-term outcomes.

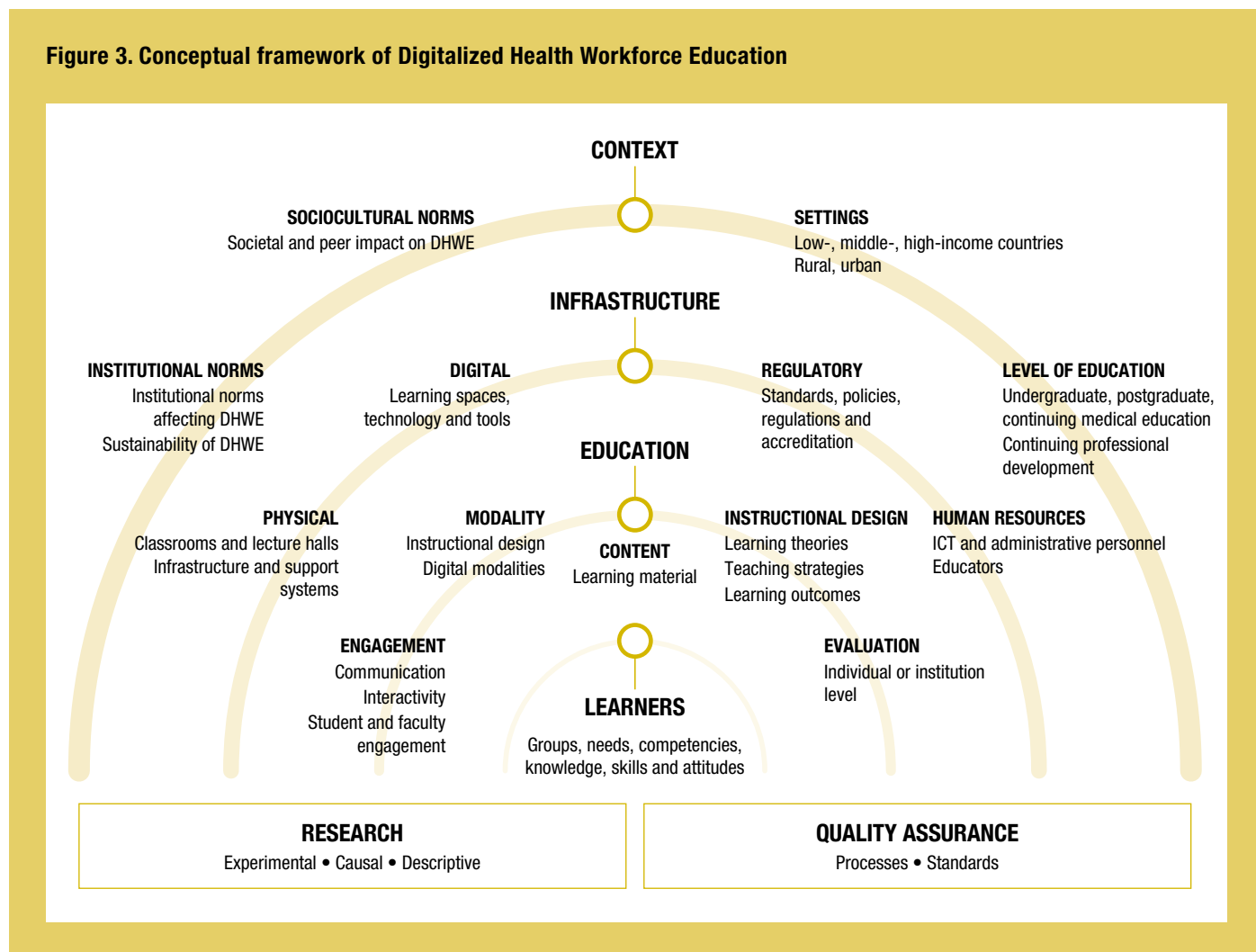
3.1 Classifying research questions from the included systematic reviews

The analysis yielded 366 concrete research questions posed in the 145 articles. The research questions were then further refined by using consistent terminology, removing duplicates as much as possible and mapping questions to similar topics. Evidence mapping resulted in the identification of a final list of 63 non-prioritized research questions covering 14 domain elements of the applied conceptual framework (Fig. 3 and Table 1). Most research questions retrieved from systematic reviews addressed the effectiveness of the DHWE modality and the content (learning material) provided within the digital platform. Additionally, some reviews addressed other subdomains such as regulations and standards (Infrastructure: regulatory), available technologies for DHWE implementation (Infrastructure: digital), and the costs and adoption of DHWE (Context: sociocultural norms, Context: institutional norms).

3.2 Applying a conceptual framework for digitalized health workforce education

The conceptual framework applied for the classification of the identified research questions outlined in this report was developed from a review and synthesis of relevant DHWE frameworks, including those focused on implementing or adopting DHWE or technology-enhanced learning in any context, as well as those focused on innovations in the field. These include the Global e-Learning framework,²¹ the e-Learning Theoretical Framework,²² the Conceptual Framework for e-Learning in Developing Countries²³ and the Innovation Adoption review.²⁴ The elements of the conceptual framework’s domains are arranged according to key themes and categories (*Context, Infrastructure, Education, Learners, Research, Quality Assurance*) schematically represented in concentric semicircles (Fig. 3). The elements captured are not intended to be exhaustive although they provide appendages for the inclusion of subcomponents. The operational description for each framework component is outlined in Annex 2.

Figure 3. Conceptual framework of Digitalized Health Workforce Education



DHWE: digitalized health workforce education; ICT: information and communication technologies.

Source: Tudor Car L, Poon S, Kyaw BM, Cook DA, Ward V, Atun R et al. Digital education for health professionals: An evidence map, conceptual framework, and research agenda. *J Med Internet Res*. 2022;24(3):e31977.

Table 1. Non-prioritized list of research questions identified in the included systematic reviews

No.	Research questions identified from systematic reviews	Conceptual framework domain and subdomain
1	How do cost and cost-related outcomes influence the adoption of DHWE?	CONTEXT: Sociocultural norms
2	How can policy-makers be supported to adopt or incorporate DHWE?	
3	How do cultural factors within different countries determine the use of digital tools in the education of health workers?	
4	How does providing access to DHWE improve learning outcomes of health workers?	
5	What is the long-term cost–effectiveness of DHWE compared to traditional education for health workers?	CONTEXT: Institutional norms
6	How does DHWE impact individual and health service outcomes, and organizational practice?	
7	Is DHWE more time–efficient than traditional education?	CONTEXT: Settings
8	What is the feasibility of implementing DHWE tools or platforms in different socioeconomic settings?	
9	What are the short- and long-term effects of using DHWE in different socioeconomic settings?	
10	Is DHWE effective in different socioeconomic settings?	
11	What are the resource requirements to implement DHWE in different socioeconomic settings?	
12	What are the challenges of implementing DHWE in different socioeconomic settings?	
13	What is the differential impact of DHWE on the clinical performance of trainee or expert surgeons?	CONTEXT: Level of education
14	How can DHWE be integrated into routine practice?	
15	How can digital technology be incorporated into current health worker education curricula to improve learning outcomes?	
16	Is DHWE effective in improving health worker competence in the clinical setting?	
17	Which features of DHWE (e.g. technical, fidelity, safety, adaptability) affect learning outcomes the most?	INFRASTRUCTURE: Digital
18	What are the minimum technology requirements needed to achieve effective DHWE?	
19	What are the technical resources needed to deliver DHWE?	
20	How should [the infrastructure of institutions] delivering DHWE be assessed or accredited?	INFRASTRUCTURE: Regulatory
21	What are the best practices for development, evaluation and use of DHWE?	
22	Is the use of accreditation-related milestones in DHWE effective?	
23	In which digital skills should educators facilitating DHWE be competent?	CONTEXT: Sociocultural norms
24	How does the digital competence of educators impact learning outcomes from DHWE?	
25	What are the human workforce resources needed for DHWE?	
26	What type of instructional design is most effective for DHWE?	

DHWE – digitalized health workforce education.

No.	Research questions identified from systematic reviews	Conceptual framework domain and subdomain
27	Which engagement components of DHWE (e.g. interactivity, feedback) contribute the most to enhanced learning outcomes?	CONTEXT: Institutional norms
28	What is the optimal use of video-assisted debriefing for health worker simulation-based training?	
29	How does the curricula design of DHWE interventions (e.g. format and modality) affect learning outcomes?	CONTEXT: Settings
30	How best can digital simulation-based training be used to train non-technical skills in health workers?	
31	What is the effectiveness of DHWE (mixed or single modality) compared to traditional methods in delivering health worker education?	
32	How effectively can DHWE complement (i.e. blend) or substitute traditional education for health workers?	
33	What benefits do digital simulation-based psychomotor skills training provide to the medical trainee?	
34	What are the barriers to obtaining DHWE materials and how can they be overcome?	
35	Following simulation-based education, what content should be included in debriefing (e.g. digital data) to achieve improved clinical outcomes?	CONTEXT: Level of education
36	Can DHWE be used to overcome challenges in delivering content-specific topics for health workers (e.g. surgical training in rare pathologic states)?	
37	What types of key learning content must be included in DHWE interventions to improve clinical competencies?	
38	How best can DHWE be designed to achieve learning outcomes denoted in the higher levels of the Kirkpatrick model? ^a	EDUCATION: Instructional design
39	What learning theories can be used to inform development of effective DHWE?	
40	Is mastery learning via DHWE more, less, or as effective as traditional education in terms of clinical psychomotor skills improvement?	
41	Is spacing digital simulation-based training more, less, or as effective as traditional education in clinical psychomotor skills development?	
42	How does the frequency and duration of digital simulation-based psychomotor skills training impact health workers' skills transfer to the clinical setting?	
43	What is the optimal duration, frequency and intensity needed for DHWE programmes to impact learning and clinical outcomes?	
44	What pedagogy should be used in DHWE to improve health workers knowledge and skills?	
45	What is the effectiveness of using DHWE to train and assess non-technical skills in health workers?	
46	What is the effectiveness of problem-based learning in DHWE?	
47	How does interactivity of DHWE programmes affect learning and clinical outcomes?	EDUCATION: Engagement
48	What is the minimum level of haptic feedback required in digital simulation-based training programmes to improve health workers' psychomotor skills?	
49	What are the levels of learner acceptability of DHWE interactivity?	

^a The Kirkpatrick Model (<https://kirkpatrickpartners.com/the-kirkpatrick-model/>)

4. Case studies

The six case studies illustrated in this section are intended to support growing evidence of positive approaches to support research implementation for DHWE. They represent good examples of initiatives led by countries and their partners that highlight best practices and outline critical success factors to enhance research and development capacity to promote DHWE.

4.1 Government leadership helps to improve the quality of maternal and newborn care through the integration of an mLearning tool in Ethiopia and India

Deliberate efforts by governments in collaboration with civil society organizations and specialized agencies can yield results to advance digital health. In this case study the Maternity Foundation, an international nongovernmental organization, worked with the Governments of Ethiopia and India to integrate the Safe Delivery App (SDA) into national basic emergency obstetrics and newborn care (BEmONC), and the quality improvement, nursing and midwifery curricula.

The SDA is a smartphone application that provides skilled birth attendants with direct and instant access to evidence-based and up-to-date clinical guidelines aligned with WHO guidelines on BEmONC. To test whether the SDA could be used to improve newborn health outcomes and the competencies of midwives, a pragmatic, cluster randomized controlled trial²⁵ was conducted with support from the University of Copenhagen and the University of Southern Denmark in five rural districts in West Wollega, Ethiopia in 2013–2014. The results showed the App to be an effective tool to improve and sustain health workers' knowledge and skills in post-partum haemorrhage and neonatal resuscitation for as long as 12 months after its introduction, and there was a positive attitude among health workers to using it. Based on the encouraging research findings, the SDA was integrated into midwifery trainings delivered across 28 health facilities in Ethiopia.

To support further roll-out of the SDA within national initiatives, the implementation model of integrating the SDA into in-service training in BEmONC was further developed and deployed. The new national BEmONC curriculum featuring the integrated 12-day training with the SDA as a teaching aid has been used in Ethiopia since mid-2018. Through these experiences of rolling out the SDA as a stand-alone aid, or through integration, feedback on implementation and evidence on learning effectiveness have been captured to improve the App and skilled birth attendants' learning outcomes.

In India, the SDA's integration into the Labour Room and Quality Improvement Initiative (LaQshya) paved the way for its integration into the Midwifery Programme, which further supported its inclusion in the BSc Nursing Curriculum, making the application an educational tool for skilled birth attendants throughout their entire service journey. The Indian Nursing Council also approved the SDA as a tool for Continued Professional Development with effect from November 2021. Throughout the institutionalization and scaling up of the App in India, some key learnings emerged:

- **Government readiness to use digital tools and a clear vision of “how” and “where” are critical.** Leadership at the highest level in the Ministry of Health and the Department of Health at the state level believed in the power of information technology, was actively involved in the roll-out of the App and had a clear vision of how it complements existing structures, systems and programmes.
- **National adaptation and integration was a prerequisite.** For the Indian version of the App to be available as a complete tool and readily available for integration, co-development and approval were necessary.
- **Ownership and partnerships should be promoted.** A flexible roll-out approach enabled states to adapt to their specific context and priorities, which increased the state-level ownership.

- **Data insights should be used.** Continuous monitoring and feedback are needed for effective and sustained roll-out. Extracting and sharing data insights with state governments and partners proved helpful to maintain the motivation for continued roll-out efforts.
- **Training should focus on the App.** Ensuring end-users understand how the tool works and the content within led to effective use of the initiative.

4.2 E-learning strategies for training and knowledge-building to advance nursing practices in Mexico

Nurses with advanced patient management skills are in short supply in Mexico. To address this shortfall, the Better Health Programme Mexico (BHPMx), a collaboration between institutions in Mexico and the United Kingdom, aims to advance knowledge and skills among nurses in the country. This collaboration allows for the exchange of knowledge and best practices to improve nursing education practices, as well as the facilitation of opportunities for professional development in nurse training.²⁶ Four pillars support the goal of empowering and educating nurses: (i) diagnosis and planning; (ii) knowledge exchange; (iii) education and training; and (iv) communication and advocacy.

Three e-learning modules are proposed within the education and training pillar: digital skills; leadership and teamwork in order to develop digital competencies; and the ability to use technology when providing health services. It is important to consider the feasibility of implementing DHWE in low-income countries, such as the availability of a reliable internet connection and training costs. Taking these factors into account, the BHPMx e-learning modules are free to use, can be used offline on smartphones and only require an internet connection to download educational materials.²⁷ This case study highlights the significance of assessing and implementing health worker education technologies that are appropriate for the specific health setting and its workers. Furthermore, it displays the viability and adaptability of environments in which DHWE can be implemented despite accessibility barriers.

4.3 Significance of government innovation and the adoption of digitalized health workforce education in the United Arab Emirates

The Dubai Health Authority (DHA) attempts to advance its health-care systems by utilizing technology to improve health worker education. To achieve this, the DHA's Medical Education and Research Department has created two digital training platforms to provide health workers with accessible medical education and academic support. Implementation of these DHWE systems hopes to increase health workers' competency in provision of quality patient care.²⁸

The first platform is known as DARB, a comprehensive digital educational system that increases resident doctors' medical expertise and knowledge.²⁸ Additionally, it provides an avenue for students to communicate with their instructors about course content and assignments. DARB currently consists of five health-care specializations and serves 60% of the medical residents. The DHA plans to update the platform to extend its applicability to other specializations as well as for use in fellowships and internships.

The second programme – MANAAR – is an interactive online learning management platform for in-service medical professionals. This digital platform will be used for ongoing professional development to assess and evaluate the performance of health workers. The system will then utilize this data to identify knowledge gaps and suggest areas where additional learning may be required.

Accredited training programmes will also be available, providing more opportunity for continuing medical education. This case study demonstrates a government's commitment and approach to innovate health worker education to improve their digital and medical competencies and patient health outcomes.

4.4 Transnational collaboration for digitalized health workforce education transformation in Denmark, Italy, Slovenia and Spain

The European Union (EU) is funding DELIVER (Digital Educational programme involVing hEalth pROfessionals) to meet the growing demand for digital competencies in health workers across the region. This digital education programme is a three-year project involving four EU countries: Denmark, Italy, Slovenia and Spain. It targets health workers and managers from various disciplines in order to increase their ability to provide telemedicine and support patients' use of health technologies.²⁹ Other goals of this project include using information and communication technologies to promote digital competency training, implementing digital technologies in health-care delivery and optimizing health worker effectiveness and efficiency.³⁰

The project involves the following components.

- 1 Needs Analysis Report.** Assess digital competencies of health workers, attitudes towards digital technologies in health care, and existing frameworks/practices for digital health.
- 2 Digital and Organizational Analysis Report.** Assess organizational readiness for digital technology implementation.
- 3 Educational Programme and e-Learning Tools:** Develop an accredited curriculum and teaching materials for educators to use in providing education to health workers through digital means.
- 4 Digital Skills Platform.** Develop a knowledge-sharing platform for open access to e-learning material for health workers and e-teaching tools for health educators.
- 5 Recommendations for Policy and Implementation.** Summarize best practices and future directions for digital education technologies in health care.

Each collaborating country will lead a work package corresponding to one of the components. The synergy of these efforts will allow DELIVER to be used as a framework for other countries/regions to follow in order to implement the digitalization of health worker education in and across nations.

4.5 Highlighting the role of investment and partnerships for advancing digitalized health workforce education in Australia

Building on the sixth pillar of its 2018 National Digital Health Strategy, the Australian Government recently developed a National Digital Health Workforce and Education Roadmap³¹ to ensure that the health workforce has the skills to use technology and digital health tools to provide the best health care to people across the country. The roadmap includes practical steps to achieve the National Digital Health Strategy's education and training goals. It is founded on a set of guiding principles, priorities, prerequisites and targets

for stakeholders such as universities and education institutions, researchers, industry, clinicians, other health-care professionals and carers. The roadmap proposes a three-horizon multipronged approach based on the progressive implementation of the following health professions priorities: (i) digital health literacy; (ii) training pathways for new and emerging roles; and (iii) governance, risk, and ethics training to support new care models.

The roadmap promotes new ideas while acknowledging the expansion of existing initiatives aimed at strengthening DHWE in the country. It provides a common framework to leverage and measure benefits, as well as case study examples to guide users and show progress. The Digital Health Cooperative Research Centre, an AUD 230 million, seven-year programme of work funded by industry and the Australian Federal Government to shape the future of digital health, is notable among the initiatives. The Centre focuses on four broad flagship areas to strengthen digital health, with an ambitious goal of recruiting more than 100 PhDs to help shape the future digital workforce during its roll-out. This will be achieved through the development of PhD education programmes, the formation of communities of practice in key areas, and the availability of capacity-building resources to support workforce skills in digital health.

The programme involves a relatively large number and variety of partners, including state and federal agencies, primary health networks, private and public services, technology companies, payers, regulators and several universities. Investments and collaborations with key stakeholders are critical to accelerate the advancement of DHWE. This case study demonstrates how clear policies and strategies, combined with the presence of active stakeholders, can inspire real progress toward this goal.

4.6 High-level action and political will promotes digitalized health workforce education in Singapore

This case study of Research, Innovation and Enterprise Council (RIEC) in Singapore shows that high-level action and political will are important prerequisites to develop and implement research agendas that positively impact DHWE. The RIEC, led by the Prime Minister's Office,³² is heavily investing in the digital sector with the goal of capitalizing on the potential of new technologies and smart solutions in knowledge-based and service-oriented industries. The development of new technologies is also transforming the design and delivery of health services and health worker education. These developments can be contextualized and applied across health sectors globally to improve pedagogies and digital health services.

RIEC conducts periodic reviews of its research outcomes to identify both successes and gaps in order to anticipate and decide where resources for further development should be allocated. This exercise, which is essentially a sharpening of its own research agenda, provides RIEC with reference areas to target for maximum development in the digital health sector. The Health Industry Transformation Map, announced in 2017, outlined strategies to move the health-care industry and workforce towards better skills and quality health-care services for Singaporeans. As distance and distributed learning becomes more commonplace, lessons from the RIEC approach can help other countries enable their research agencies and authorities to improve health workforce skills by promoting and supporting flexible, user-friendly, self-paced, lifelong and cost-effective training delivered through equitable digital platforms.

5. Discussion

Analysis of the literature gathered for this report revealed 10 broad digital modalities from a range of topics for health worker education. It should also be noted that the evidence analysed for this paper relates broadly to the main health-care occupations, and thus caution must be applied in wider interpretation (to mid-level and other occupational groups) of the findings or recommendations. While most studies reported an equivalent or greater positive effect on learning outcomes when compared to traditional delivery modes, evidence of effectiveness was generally of low or very low quality, had an unclear or high risk of bias and had other methodological limitations.

Similarly, the majority of included studies only reported immediate, short-term results, which is inevitable in situations where the learning programmes were short. However, levels of effectiveness should be compared differently when applied to longer periods of learning, whether pre- or in-service. This highlights the need for additional studies using robust methods to examine the long-term effectiveness of DHWE.

Given the COVID-19 pandemic's rapid disruption of traditional education, many education providers are increasingly relying on digital tools and platforms. While reviewed studies showed that DHWE increased learners' efficacy and knowledge, the quality of evidence was found to be variable as the risk of bias from individual studies was high. More studies are therefore needed using randomized controlled trials or experimental techniques to test the effectiveness of DHWE interventions for pre- and in-service training. Also, more studies describing the exponential acceleration of learning technology adoption and use are needed to provide insights into which interventions are effective in both acute and non-emergency situations.

To illustrate actual examples of how DHWE is being implemented and adopted, case studies from different regions are provided in this report. Domains and subdomains from the conceptual framework were also mapped to the case studies to test its applicability. Specifically, the conceptual framework provides a practical approach for considerations to be made when investing time, effort and financial resources in implementing DHWE. Some of the case studies described sought to advance the acceptability, feasibility and adoption of DHWE in their respective countries, institutions and settings (see Fig. 3, Context: sociocultural norms, institutional norms, settings). Others considered the infrastructure, such as whether current information and communication systems are adequate to support the execution of DHWE (Fig. 3, Infrastructure: digital) or whether the regulations and standards of DHWE delivery would be met (Fig. 3, Infrastructure: regulatory). Intentional consideration of these conceptual framework domains and subdomains can serve as a guide to real-world action in meeting the goal of nurturing education in health workers and improving their competence levels.

5.1 Evidence gaps

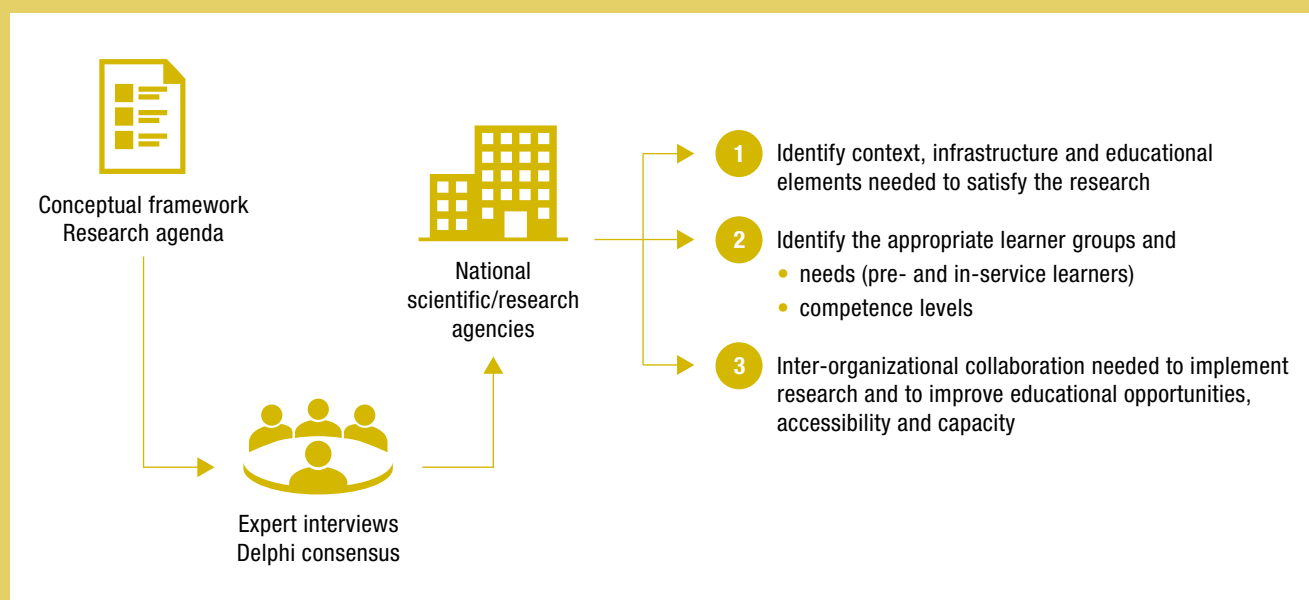
Gaps in current evidence from the literature were also identified related to various categories, modalities, participants and settings of DHWE. Most reviews compared the effectiveness of DHWE to non-digital or traditional modalities. As such, there is a lack of evidence comparing the effectiveness of one digital modality to another, making it challenging to determine how and when DHWE should be implemented. The research questions posed in the case studies are framed to be applicable to a wide range of digital modalities, though some may be more applicable to a specific digital modality. It is recommended that the research questions be contextualized so that they are appropriate for the geographic region or setting in which health worker education is delivered digitally. As the field is still in its infancy, the research questions can also serve as a guide for developing future research areas or topics.

The current evidence base on DHWE in low-income countries is limited, with a lack of primary research from such settings. This highlights an evidence gap in understanding the barriers to implementing and adopting more appropriate DHWE in low-resource settings, where training and retaining competent educators for health professionals can be challenging. Urgent, relevant, and prioritized research is needed to establish a sound evidence base, particularly given the realistic prospect that DHWE could play in these settings.

5.2. Institutional pathways and operationalizing research for digitalized health workforce education

While it's important to consider the specific context of a study when deciding which research questions to pursue, it's also crucial to aim for generalizability in order to maximize the impact of research findings. Researchers are encouraged to use the list of research questions provided in Table 1 as a reference source for developing their own questions, with the aim of identifying factors that are transferable across different contexts. This approach could involve running the DHWE programme in various settings or identifying the key features between settings that affect the learning outcomes. In addition to considering contextual factors, researchers should also incorporate objectives related to equity and gender issues, as these are important to ensuring that DHWE is accessible and effective for all learners.

Figure 4. Institutional pathways to analyse research gaps for digitalized health workforce education



The set of research questions in table 1 is relevant to current and future research proposals to strengthen the education of health workers using digital tools. Caution must be applied, however, as the research questions outlined in this brief have not been prioritized and are not exhaustive. These limitations are evident in the development of previous research agendas that incorporate further layers of Delphi studies and stakeholder inputs to elicit further questions, validate relevance and prioritization. For policy-makers, researchers and other users, several methods are available to determine with greater relevance which questions are the most important per context. These include further subjecting identified priority research topics to the Delphi approach,³³ the Child Health and Nutrition Research Initiative (CHNRI)³⁴ and Essential National Health Research and Priority Setting of the Council of Health Research for Development (COHRED).³⁵ WHO has used a combination of these methodologies^{36–39} in regional and global studies to determine priorities in other areas of health services research, which are beyond the scope of this report. This document can serve as a starting basis, providing a framework for guiding research questions to achieve the broader goals of strengthening DHWE. Although it is important to prioritize generalizable research questions, it is also critical to recognize the unique contexts and needs of different settings. Researchers could adapt an approach that mirrors or borrows from the following pathways:

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| 1 gather all stakeholders and conduct a comprehensive need assessment of DHWE to identify priority research topics; | 2 assess needs to identify the most relevant priority levels according to context using consensual methods; | 3 refer to lessons learnt from case studies of DHWE initiatives and look for factors that promote scaling up and sustainability; ^{40,41} | 4 map and procure the resources needed, including but not limited to human, financial, infrastructure and technology. |
|--|--|--|--|

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Annex 1.

List of systematic reviews informing research questions

No.	Year	Author(s)	Title	Journal
1	2014	George et al.	Online eLearning for undergraduates in health professions: A systematic review of the impact on knowledge, skills, attitudes and satisfaction	J Glob Health
2	2014	Lahti et al.	Impact of e-learning on nurses' and student nurses knowledge, skills, and satisfaction: a systematic review and meta-analysis	Int J Nurs Stud
3	2014	Mickan et al.	Use of handheld computers in clinical practice: a systematic review	BMC Med Inform Decis Mak
4	2014	Qiao et al.	The effect of virtual endoscopy simulator training on novices: a systematic review	PLoS One
5	2014	Rasmussen et al.	Offline eLearning for undergraduates in health professions: a systematic review of the impact on knowledge, skills, attitudes and satisfaction	J Glob Health
6	2014	Thepwongsa et al.	Online continuing medical education (CME) for GPs: does it work? A systematic review	Aust Fam Physician
7	2015	Cheng et al.	The use of high-fidelity manikins for advanced life support training – a systematic review and meta-analysis	Resuscitation
8	2015	Jayakumar et al.	e-Learning in surgical education: a systematic review	J Surg Educ
9	2015	Osborne et al.	The effectiveness of the use of high fidelity simulators in obstetric ultrasound training: a systematic review	Austras J Ultrasound Med
10	2015	Piromchai et al.	Virtual reality training for improving the skills needed for performing surgery of the ear, nose or throat (Review)	Cochrane Database Syst Rev
11	2015	Kato et al.	Literature review of simulation training for acute obstetric emergencies	J Jpn Acad Midwif
12	2016	Aïm et al.	Effectiveness of virtual reality training in orthopaedic surgery	Arthroscopy
13	2016	Alaker et al.	Virtual reality training in laparoscopic surgery: a systematic review & meta-analysis	Int J Surg
14	2016	Boutefnouchet et al.	Transfer of arthroscopic skills from computer simulation training to the operating theatre: a review of evidence from two randomized controlled studies	SICOT J
15	2016	den Harder et al.	The importance of human—computer interaction in radiology e-learning	J Digit Imaging
16	2016	Glassman et al.	Effect of playing video games on laparoscopic skills performance: a systematic review	J Endourol
17	2016	Liu et al.	The effectiveness of blended learning in health professions: systematic review and meta-analysis	J Med Internet Res
18	2016	Maertens et al.	Systematic review of e-learning for surgical training	Br J Surg
19	2016	Mok et al.	Effectiveness of high-fidelity patient simulation in teaching clinical reasoning skills	Clin Simul Nurs
20	2016	Santos et al.	Effectiveness of e-learning in oral radiology education: a systematic review	J Dent Educ
21	2016	Sinclair et al.	The effectiveness of internet-based e-learning on clinician behaviour and patient outcomes: a systematic review	Int J Nurs Stud
22	2016	Tarpada et al.	E-learning in orthopedic surgery training: a systematic review	J Orthop
23	2017	Lui et al.	Evaluating the effect of virtual reality temporal bone simulation on mastoidectomy performance: a meta-analysis	Otolaryngol Head Neck Surg
24	2017	Pfandler et al.	Virtual reality-based simulators for spine surgery: a systematic review	Spine J

No.	Year	Author(s)	Title	Journal
25	2017	Richmond et al.	A systematic review and meta-analysis of online versus alternative methods for training licensed health care professionals to deliver clinical interventions	BMC Med Educ
26	2017	Tomesko et al.	The effectiveness of computer-assisted instruction to teach physical examination to students and trainees in the health sciences professions: a systematic review and meta-analysis	J Med Educ Curric Dev
27	2017	Rohwer et al.	E-learning of evidence-based health care (EBHC) to increase EBHC competencies in healthcare professionals: a systematic review	Campbell Systematic Reviews
28	2017	Voutilainen et al.	Conventional vs. e-learning in nursing education: a systematic review and meta-analysis	Nurse Educ Today
29	2018	Ahmet et al.	Is video-based education an effective method in surgical education? A systematic review	J Surg Educ
30	2018	Gorbanov et al.	A systematic review of serious games in medical education: quality of evidence and pedagogical strategy	Med Educ Online
31	2018	Kang et al.	Evaluating web-based nursing education's effects: a systematic review and meta-analysis	West J Nurs Res
32	2018	Khan et al.	Virtual reality simulation training for health professions trainees in gastrointestinal endoscopy (Review)	Cochrane Database Syst Rev
33	2018	Koohestani et al.	The educational effects of mobile learning on students of medical sciences: a systematic review in experimental studies	J Adv Med Educ Prof
34	2018	Lee et al.	Mobile technology in undergraduate nursing education: a systematic review	Healthc Inform Res
35	2018	Lioffi et al.	The effectiveness of online pain resources for health professionals: a systematic review with subset meta-analysis of educational intervention studies	Pain
36	2018	Ramarat et al.	Using simulation to teach echocardiography: a systematic review	Simul Healthc
37	2018	Vaona et al.	E-learning for health professionals (Review)	Cochrane Database Syst Rev
38	2018	Zhou et al.	[Effectiveness of mobile learning in medical education: a systematic review]	Nan fang yi ke da xue xue bao (Journal of Southern Medical University)
39	2019	Botelho et al.	A systematic review of e-learning outcomes in undergraduate dental radiology curricula – levels of learning and implications for researchers and curriculum planners	Dentomaxillofac Radiol
40	2019	Brusamento et al.	Digital health professions education in the field of pediatrics: systematic review and meta-analysis by the Digital Health Education Collaboration	J Med Internet Res
41	2019	Campbell et al.	Effectiveness of online cancer education for nurses and allied health professionals: a systematic review using Kirkpatrick Evaluation Framework	J Cancer Educ
42	2019	Divakar et al.	Digital education of health professionals on management of domestic violence: systematic review and meta-analysis by the Digital Health Education Collaboration	J Med Internet Res
43	2019	Dunleavy et al.	Mobile digital education for health professions: systematic review and meta-analysis by the Digital Health Education Collaboration	J Med Internet Res
44	2019	Gentry et al.	Serious gaming and gamification education in health professions: systematic review	J Med Internet Res
45	2019	George et al.	Online digital education for postregistration training of medical doctors: systematic review by the Digital Health Education Collaboration	J Med Internet Res

No.	Year	Author(s)	Title	Journal
46	2019	Guedes et al.	Virtual reality simulator versus box-trainer to teach minimally invasive procedures: a meta-analysis	Int J Surg
47	2019	Huang et al.	Digital health professions education on diabetes management: systematic review by the Digital Health Education Collaboration	J Med Internet Res
48	2019	Kim et al.	Effect of smartphone-based mobile learning in nursing education: a systematic review and meta-analysis	Asian Nurs Res
49	2019	Kononowicz et al.	Virtual patient simulations in health professions education: systematic review and meta-analysis by the Digital Health Education Collaboration	J Med Internet Res
50	2019	Kyaw et al.	Health professions digital education on antibiotic management: systematic review and meta-analysis by the Digital Health Education Collaboration	J Med Internet Res
51	2019	Kyaw et al.	Effectiveness of digital education on communication skills among medical students: systematic review and meta-analysis by the Digital Health Education Collaboration	J Med Internet Res
52	2019	Kyaw et al.	Offline digital education for medical students: systematic review and meta-analysis by the Digital Health Education Collaboration	J Med Internet Res
53	2019	Kyaw et al.	Virtual reality for health professions education: systematic review and meta-analysis by the Digital Health Education Collaboration	J Med Internet Res
54	2019	La Cerra et al.	Effects of high-fidelity simulation based on life-threatening clinical condition scenarios on learning outcomes of undergraduate and postgraduate nursing students: a systematic review and meta-analysis	BMJ Open
55	2019	Li et al.	The effects of blended learning on knowledge, skills, and satisfaction in nursing students: a meta-analysis	Nurse Educ Today
56	2019	Nagendrababu et al.	Effectiveness of technology-enhanced learning in endodontic education: a systematic review and meta-analysis	Int Endod J
57	2019	Pei et al.	Does online learning work better than offline learning in undergraduate medical education? A systematic review and meta-analysis	Med Educ Online
58	2019	Posadzki et al.	Offline digital education for postregistration health professions: systematic review and meta-analysis by the Digital Health Education Collaboration	J Med Internet Res
59	2019	Semwal et al.	Digital education for health professions on smoking cessation management: systematic review by the Digital Health Education Collaboration	J Med Internet Res
60	2019	Tudor Car et al.	Digital problem-based learning in health professions: systematic review and meta-analysis by the Digital Health Education Collaboration	J Med Internet Res
61	2019	Tudor Car et al.	Health professions digital education on clinical practice guidelines: a systematic review by Digital Health Education Collaboration	BMC Med
62	2019	Viljoen et al.	Is computer-assisted instruction more effective than other educational methods in achieving ECG competence amongst medical students and residents? A systematic review and meta-analysis	BMJ Open
63	2019	Vitale et al.	The value of virtual reality simulators in hysteroscopy and training capacity: a systematic review	Minim Invasive Ther Allied Technol
64	2019	Wahabi et al.	Medical doctors' offline computer-assisted digital education: systematic review by the Digital Health Education Collaboration	J Med Internet Res

No.	Year	Author(s)	Title	Journal
65	2019	Xu et al.	Digital education for health professions in the field of dermatology: a systematic review by Digital Health Education Collaboration	Acta Derm Venereol
66	2019	Zhang et al.	Effectiveness of video-assisted debriefing in simulation-based health professions education: a systematic review of quantitative evidence	Nurse Educ
67	2019	Lima et al.	Effectiveness of the distance learning strategy applied to orthodontics education: a systematic literature review	Telemed J E Health
68	2020	Barteit et al.	Evaluation of e-learning for medical education in low- and middle-income countries: a systematic review	Comput Educ
69	2020	Berry et al.	Effectiveness of technology-enhanced learning to improve periodontics educational outcomes: a systematic review	J Dent Educ
70	2020	Dromey et al.	A systematic review and meta-analysis of the use of high-fidelity simulation in obstetric ultrasound	Simul Healthc
71	2020	Foronda et al.	Virtual simulation in nursing education: a systematic review spanning 1996 to 2018	Simul Healthc
72	2020	Lapierre et al.	Effectiveness of interprofessional manikin-based simulation training on teamwork among real teams during trauma resuscitation in adult emergency departments: a systematic review	Simul Healthc
73	2020	Lee et al.	Effective virtual patient simulators for medical communication training: a systematic review	Med Educ
74	2020	Männistö et al.	Digital collaborative learning in nursing education: a systematic review	Scand J Caring Sci
75	2020	Martinengo et al.	Digital health professions education on chronic wound management: a systematic review	Int J Nurs Stud
76	2020	Rangarajan et al.	Systematic review of virtual haptics in surgical simulation: a valid educational tool?	J Surg Educ
77	2020	Rothschild et al.	Does virtual reality simulation training result in fewer real-life cataract surgery complications? A systematic literature review	J Cataract Refract Surg
78	2020	Rourke	How does virtual reality simulation compare to simulated practice in the acquisition of clinical psychomotor skills for pre-registration student nurses? A systematic review	Int J Nurs Stud
79	2020	Chen et al.	Effectiveness of virtual reality in nursing education: meta-analysis	J Med Internet Res
80	2020	Hanshaw et al.	High fidelity simulation evaluation studies in nursing education: a review of the literature	Nurse Educ Pract
81	2020	Khalaf et al.	Effectiveness of technology-enhanced teaching and assessment methods of undergraduate preclinical dental skills: a systematic review of randomized controlled clinical trials	BMC Med Educ
82	2020	Portelli et al.	Virtual reality training compared with apprenticeship training in laparoscopic surgery: a meta-analysis	Ann R Coll Surg Engl
83	2020	Richardson et al.	Virtual patient technology to educate pharmacists and pharmacy students on patient communication: a systematic review	BMJ Simul Technol Enhanc Learn
84	2020	Srinivasa et al.	The role of online videos in teaching procedural skills to post-graduate medical learners: a systematic narrative review	Med Teach

No.	Year	Author(s)	Title	Journal
85	2020	Vallée et al.	Blended learning compared to traditional learning in medical education: systematic review and meta-analysis	J Med Internet Res
86	2020	Zhao et al.	The effectiveness of virtual reality-based technology on anatomy teaching: a meta-analysis of randomized controlled studies	BMC Med Educ
87	2021	Al Asmri et al.	Effectiveness of technology-enhanced simulation in teaching digital rectal examination: a systematic review narrative synthesis	BMJ Simul Technol Enhanc Learn
88	2021	Balakrishnan et al.	Effectiveness of blended learning in pharmacy education: a systematic review and meta-analysis	PLoS One
89	2021	Berg et al.	Effectiveness of online communication skills training for cancer and palliative care health professionals: a systematic review	Psychooncology
90	2021	Brown et al.	The effect of high-fidelity simulation-based learning in acute cardiorespiratory physical therapy – a mixed-methods systematic review	J Phys Ther Educ
91	2021	Cabral et al.	The use of information and communication technology in continuing education in tuberculosis	J Eur CME
92	2021	Capitani et al.	The role of virtual reality in knee arthroscopic simulation: a systematic review	Musculoskelet Surg
93	2021	Chae et al.	Effectiveness of virtual simulation to enhance cultural competence in pre-licensure and licensed health professionals: a systematic review	Clin Simul Nurs
94	2021	Chen et al.	Effects of mobile learning for nursing students in clinical education: a meta-analysis	Nurse Educ Today
95	2021	Clarke et al.	Virtual reality simulation – the future of orthopaedic training? A systematic review and narrative analysis	Adv Simul
96	2021	Co et al.	Distance education for anatomy and surgical training – a systematic review	Surgeon
97	2021	Gao et al.	Comparison of the influence of massive open online courses and traditional teaching methods in medical education in China: a meta-analysis	Biochem Mol Biol Educ
98	2021	Gupta et al.	Can video games enhance surgical skills acquisition for medical students? A systematic review	Surgery
99	2021	Gelmini et al.	Virtual reality in interventional radiology education: a systematic review	Radiol Bras
100	2021	He et al.	Synchronous distance education vs traditional education for health science students: a systematic review and meta-analysis	Med Educ
101	2021	Jhou et al.	Different pedagogies for acquisition of knowledge and skill: a systematic review and network meta-analysis	Postgrad Med J
102	2021	Kovoor et al.	Validity and effectiveness of augmented reality in surgical education: a systematic review	Surgery
103	2021	Lactona et al.	Efficacy and knowledge of conducting CPR through online learning during the COVID-19 pandemic: A literature review	J Public Health Res
104	2021	Lin et al.	Virtual reality training for cataract surgery operating performance in ophthalmology trainees	Cochrane Database Syst Rev
105	2021	Lowe et al.	Immersive technology in ophthalmology education: a systematic review	BMJ Simul Technol Enhanc Learn
106	2021	Maheu-Cadotte et al.	Efficacy of serious games in healthcare professions education: a systematic review and meta-analysis	Simul Healthc

No.	Year	Author(s)	Title	Journal
107	2021	Mao et al.	Immersive virtual reality for surgical training: a systematic review	J Surg Res
108	2021	Moro et al.	Virtual and augmented reality enhancements to medical and science student physiology and anatomy test performance: a systematic review and meta-analysis	Anat Sci Educ
109	2021	Mulyadi et al.	Effects of simulation technology-based learning on nursing students' learning outcomes: a systematic review and meta-analysis of experimental studies	Nurse Educ Today
110	2021	do Nascimento et al.	Effectiveness of the serious game for learning in nursing: systematic review	Rev Gaucha Enferm
111	2021	Ødegaard et al.	Digital learning designs in physiotherapy education: a systematic review and meta-analysis	BMC Med Educ
112	2021	Patano et al.	Education technology in orthodontics and paediatric dentistry during the COVID-19 pandemic: a systematic review	Int J Environ Res Public Health
113	2021	Rodríguez-Abad et al.	A systematic review of augmented reality in health sciences: A guide to decision-making in higher education	Int J Environ Res Public Health
114	2021	Schmidt et al.	Virtual reality simulation in robot-assisted surgery: meta-analysis of skill transfer and predictability of skill	BJS Open
115	2021	Shih et al.	Effects of digital learning in anaesthesiology: A systematic review and meta-analysis	Eur J Anaesthesiol
116	2021	Shorey et al.	The use of virtual reality simulation among nursing students and registered nurses: a systematic review	Nurse Educ Today
117	2021	Sinacori et al.	The effect of distance learning on knowledge acquisition in undergraduate second-degree nursing students: a systematic review	Nurs Educ Perspect
118	2021	Soloe et al.	The effectiveness of mHealth and eHealth tools in improving provider knowledge, confidence, and behaviors related to cancer detection, treatment, and survivorship care: a systematic review	J Cancer Educ
119	2021	Taba et al.	The development of laparoscopic skills using virtual reality simulations: a systematic review	PLoS One
120	2021	Thompson et al.	The future of orthopaedic surgical education: where do we go now?	Surgeon
121	2021	Tolarba	Virtual simulation in nursing education: a systematic review	Int J Nurs Educ
122	2021	Verville et al.	Using technology-based educational interventions to improve knowledge about clinical practice guidelines	J Chiropr Educ
123	2021	Wang et al.	Evaluating evidences of effectiveness in web-based nursing education program	Comput Math Methods Med
124	2021	Wu et al.	Perceptions of medical students towards and effectiveness of online surgical curriculum: a systematic review	BMC Med Educ
125	2021	Zhang et al.	Virtual reality simulation in training endoscopic skills: a systematic review	Laparosc Endosc Robot Surg
126	2021	Zhao et al.	The comparison of teaching efficiency between virtual reality and traditional education in medical education: a systematic review and meta-analysis	Ann Transl Med
127	2022	Chandran et al.	Mobile applications in medical education: a systematic review and meta-analysis	PLoS One
128	2022	Choi et al.	Effectiveness of immersive virtual reality in nursing education: systematic review	Nurse Educ

No.	Year	Author(s)	Title	Journal
129	2022	Clerkin et al.	What is the impact of video as a teaching method on achieving psychomotor skills in nursing? A systematic review and meta-analysis	Nurse Educ Today
130	2022	Daniel et al.	Video-based coaching for surgical residents: a systematic review and meta-analysis	Surg Endosc
131	2022	Du et al.	Blended learning vs traditional teaching: The potential of a novel teaching strategy in nursing education – a systematic review and meta-analysis	Nurse Educ Pract
132	2022	Hao et al.	Application of digital education in undergraduate nursing and medical interns during the COVID-19 pandemic: a systematic review	Nurse Educ Today
133	2022	Jallad et al.	The effectiveness of virtual reality simulation as learning strategy in the acquisition of medical skills in nursing education: a systematic review	Ir J Med Sci
134	2022	Ozdemir et al.	Game-based learning in undergraduate nursing education: a systematic review of mixed-method studies	Nurse Educ Pract
135	2022	Lei et al.	Effects of high-fidelity simulation teaching on nursing students' knowledge, professional skills, and clinical ability: A meta-analysis and systematic review	Nurse Educ Pract
136	2022	Li et al.	High-fidelity simulation in undergraduate nursing education: A meta-analysis	Nurse Educ Today
137	2022	Min et al.	Effectiveness of serious games in nurse education: a systematic review	Nurse Educ Today
138	2022	Moussa et al.	Effectiveness of virtual reality and interactive simulators on dental education outcomes: systematic review	Eur J Dent
139	2022	Plackett et al.	The effectiveness of using virtual patient educational tools to improve medical students' clinical reasoning skills: a systematic review	BMC Med Educ
140	2022	Savage et al.	E-learning in emergency medicine: a systematic review	Emerg Med Australas
141	2022	ShahAli et al.	Using e-learning methods for physiotherapy students learning – a systematic review and meta-analysis of the impact on knowledge, skills, satisfaction and attitudes	Eur J Physiother
142	2022	Sim et al.	Virtual simulation to enhance clinical reasoning in nursing: a systematic review and meta-analysis	Clin Simul Nurs
143	2022	Stake-Nilsson et al.	Medication dosage calculation among nursing students: does digital technology make a difference? A literature review	BMC Nurs
144	2022	Thangavelu et al.	Digital serious games in developing nursing clinical competence: a systematic review and meta-analysis	Nurse Educ Today
145	2022	Tursø-Finnich et al.	Virtual reality head-mounted displays in medical education – a systematic review	Simul Healthc

Annex 2.

Operational definitions for components of the conceptual framework

Domain	Category	Definition
Context	Sociocultural norms	Acceptability and adoption of digitalized health workforce education as a form and norm of education within the society.
	Institutional norms	Acceptability, impact, considerations and processes in relation to adoption of digitalized health workforce education at the institutional level.
	Settings	The setting in which digitalized health workforce education is conducted or implemented, including to low-, middle- and high-income countries as well as rural or urban environments.
	Level of education	Competency level acquired or degree status obtained and how digital technology may impact or aid further learning.
Infrastructure	Physical	The physical learning space within which the in-person component of blended digitalized health workforce education takes place.
	Digital	Information and communication technology devices (both hardware and software) to support and create learning environments (virtual environments, digital networks, technological modifications) or media for digitalized health workforce education, as well as the bandwidth available.
	Regulatory	The policies and regulatory standards for health worker licensing and accreditation, as well as those pertaining to digitalized health workforce education.
	Human resources	The human resources required for digitalized health workforce education to be maintained and sustained, including education, administrative and information technology personnel.
Education	Modality	The choice and configuration of digitalized health workforce education modality (e.g. online learning, mLearning) and its potential blending within in-person education.
	Instructional design	The method and practice of health worker education encompassing teaching strategies, learning theories and educational outcomes using digital tools.
	Content	Education area, discipline, theme or topic delivered through digital health workforce education.
	Engagement	The extent of communication, interactivity, immersion and feedback to participants receiving learning through digital means.
	Evaluation	Evaluation of digitalized health workforce education conducted at individual and/or institutional level to determine its impact on improvement in educational and clinical outcomes.
Learner	Groups, needs, competencies, knowledge, skills, attitudes	Health workers with distinctive needs, competencies (underpinned by knowledge, skills, attitudes), requirements and digital literacy levels necessary to work and learn individually or in teams.
Research	Implementation; conceptualization; novel solutions	Systematic study of the use of digitalized health workforce education with the aim of providing robust evidence to support its effective and efficient use.
Quality assurance	Processes, resources, outputs, standards	The processes, resources, outputs and standards used in the audit of digitalized health workforce education.

Source: Tudor Car L, Poon S, Kyaw BM, Cook DA, Ward V, Atun R et al. Digital education for health professionals: An evidence map, conceptual framework, and research agenda. J Med Internet Res. 2022;24(3):e31977.

Annex 3.

List of digitalized health workforce education modalities and their definitions

DHWE modality	Working definition/description
Augmented reality¹	Augmented reality, a type of mixed reality, is a real world-based experience enhanced by digital objects or information.
Massive open online course	An online course that is designed for large numbers of geographically dispersed participants.
Mobile learning (mLearning)	Learning that uses personal electronic devices, potentially across multiple contexts, through social and content interactions.
Offline digitalized health workforce education	Education delivery that requires no internet or local area network connection and can be delivered through an external media, including CD-ROM, external hard disk and USB stick.
Online digitalized health workforce education	Education delivery that requires the use of a Transmission Control Protocol and an Internet Protocol, also referred to as online, web-based or networked.
Serious gaming and gamification	Learning activities that are set within a competitive activity and promote either learning or cognitive skills development, or allow practising of psychomotor skills in a virtual environment.
Virtual reality	Real-time, interactive exploration with a computer-generated, real or artificial three-dimensional multimedia sensory environment. Users experience active learning through different levels of immersion, where they interact with objects in or through actions with the digital world.
Virtual patient	Computer simulations of clinical scenarios in real time for the purpose of medical teaching, education or assessment.
Hologram²	The choice and configuration of digital education modality (e.g. online learning, mLearning) and its potential blending within in-person education.
Digital psychomotor skills trainers	The method and practice of health worker education encompassing teaching strategies, learning theories and educational outcomes using digital tools.
Simulation-based medical training	Education area, discipline, theme or topic delivered through digital education.
Blended learning	The extent of communication, interactivity, immersion and feedback to participants receiving learning through digital means.

¹ Dhar P, Rocks T, Samarasinghe RM, Stephenson G, Smith C. Augmented reality in medical education: students' experiences and learning outcomes. *Med Educ Online*. 2021;26(1):1953953.

² Mishra S. Hologram the future of medicine – From Star Wars to clinical imaging. Indian Heart J. 2017;69(4):566–7.

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